

Exercise 4

Consider N non-interacting spin- $\frac{1}{2}$ magnetic moments μ in an external magnetic field B . The two energy levels are

$$E_{\uparrow} = -\mu B, \quad E_{\downarrow} = +\mu B.$$

- Show that the single-spin partition function is

$$Z_1 = e^{\beta\mu B} + e^{-\beta\mu B} = 2 \cosh(\beta\mu B), \quad \beta = \frac{1}{k_B T}$$

- Hence find the free energy of the whole system:

$$F = -Nk_B T \ln[2 \cosh(\beta\mu B)].$$

- Using

$$S = - \left(\frac{\partial F}{\partial T} \right)_B,$$

Exercise 5

Consider N non-interacting spin- $\frac{1}{2}$ magnetic moments μ in an external magnetic field B . The two energy levels are

$$E_{\uparrow} = -\mu B, \quad E_{\downarrow} = +\mu B.$$

- show that the entropy is

$$S(B, T) = Nk_B \left[\ln(2 \cosh x) - x \tanh x \right], \quad x = \frac{\mu B}{k_B T}.$$

- Show that S depends only on the ratio

$$x = \frac{\mu B}{k_B T}.$$

- Explain why, in an adiabatic process ($S = \text{const}$), this implies

$$\frac{B}{T} = \text{const}, \quad \text{hence} \quad T_f = T_i \frac{B_f}{B_i}.$$